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The Effects of Muscle Energy Technique on Cervical Range of Motion

The purpose of this study was to determine if a four week treatment period of muscle energy technique (MET) would significantly increase cervical flexion, extension, sidebending, and rotation in asymptomatic persons with limited range of motion (ROM). Eighteen volunteers qualified as subjects for the study following screening for neck ROM limitation. These subjects were then randomly assigned to either a control or MET group. A series of six, mixed, two-way analyses variance (ANOVA) were used to test for significant cervical ROM increases. The two factors examined were Group (MET vs. control) and Test (pre vs. post). Significant interactive effects for both left and right rotation were found (both F 's > 4.8 and p 's < 0.05) indicating a significantly greater ROM in the MET group. Trends toward significance were found for the remaining ranges of motion with the mean measures for each of the treatment groups showing an increase between pre-test and post-test. These results support MET as an effective technique for increasing cervical range of motion.

Keywords: Muscle Energy, Cervical Spine, Range of Motion

CERVICAL range of motion may be affected by several musculoskeletal conditions. The cervical spine can be subjected to acute flexion and extension injuries, the so called whiplash, as well as repetitive injury resulting from improper posture and abnormal positions of the head and neck (1). These conditions, along with muscle spasm due to joint dysfunction or irritation, and structural problems such as vertebral rotation, can have adverse effects on normal neck range of motion (2). Limitations in range of motion can lead to muscle imbalances, changes at the articulating sur-

faces, and pain. Physical therapy goals may include the restoration of normal cervical range of motion.

The muscle energy technique (MET) has been used to clinically restore range of motion in vertebral segments of the spine (3). MET as defined by Greenman is a manual medicine treatment procedure which involves voluntary contraction by the patient, against a directly executed counterforce applied by the operator (1). It is an active technique in which the patient contributes corrective muscular force and can directly affect the dosage applied by varying their amount of effort. MET can be used to lengthen muscles which are shortened, contractured or spastic. This technique can also strengthen physiologically weakened muscles and reduce localized edema through muscle pump action. Greenman reports that mobilization of any articulation with decreased mobility can be achieved with MET (1).

Greenman outlines three types of muscle contractions utilized in MET as follows:

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1. Isometric contraction-the distance between origin and insertion of the muscle is maintained at a constant length.
2. Concentric contraction-the origin and insertion approximate during contraction.
3. Eccentric contraction-muscle tension allows a controlled separation of origin and insertion with a lengthening of the muscle (1).

For the purpose of this study an isometric muscle contraction was utilized for the following reasons: (a) shortened and hypertonic muscles are a major component of restricted motion. (b) After an isometric contraction a hypertonic muscle can be passively stretched to a new resting length due to the affect of neural input on gamma control to the fusiform motor mechanism. (c) The muscle contraction stimulates firing of the golgi tendon organ causing muscle fibers to relax and lengthen (3). The key to effective treatment is that both the operator and the patient relax their counterforces simultaneously, but that the operator maintains the segment position and readjusts the segment to compensate for the changes made before repeating the procedure (7).

The barrier model (10) attempts to explain the underlying pathology of altered motion. Normal joints possess a greater passive range of motion as compared to their active range of motion. It is possible for a barrier, or motion restriction, to form in one or more directions between neutral position and the normal limit of active range of motion. The patient is then unable to achieve full active range unless an additional force is applied (6). MET has been used clinically in an attempt to overcome the restrictions to joint movement imposed by these barriers, thereby restoring normal active joint motion in the spine.

Both direct and indirect methods can be used for positioning and applying the counterforce during MET (3). In this method, the patient is positioned into the restrictive barrier and then asked to exert an isometric force away for the barrier. The amounts of force and counterforce needed are governed by the length and strength of the muscle groups involved (3). The smaller muscles of the spine in this case, required only small counterforces to create an effect.

No published empirical research was uncovered on the effectiveness of MET, although descriptive and theoretical articles have been published. There were, however, two unpublished research studies on MET, one for increasing lumbar flexion (4), and one on the efficacy of MET on mid-cervical range of motion (5).

Normal subjects were used in this study to control for any extraneous variables that may have occurred in patients at different stages of pathology such as stage of healing and pain.

The CROM was used as a control for any mea-

surement error that could have occurred with using either the standard or bubble goniometer, such as the instrument not being in the identical starting point from one measurement to the next or from one subject to another. The CROM uses the ears and nose as its anatomical anchor points, therefore, there was no difference in tool placement between pre-test and post-test measurements. This eliminated a source of error that may have been caused by the person performing the measurements.

One extraneous variable was the fact that each subject did not have limitation in the same direction of motion. The treatment group did not undergo identical treatments. Each subject was treated according to his or her own direction of limitation.

There have been no published empirical studies on the effectiveness of MET, although descriptive and theoretical articles were found. For example, Goodridge defined and described various methods of MET and stated that it could be used to treat dysfunctions of both the spine and extremities (3). Geiringer and deLateur (6) and Kimberly (7) discussed MET as a form of manipulative treatment and described the physiology behind the technique.

A thesis undertaken by Wolfson (4) examined the effectiveness of MET, and found descriptive and theoretical articles were found. For example, Goodridge (3) defined and described various methods of MET and stated that it could be used to treat dysfunctions of both the spine and extremities. Geiringer and deLateur (6) and Kimberly (7) discussed MET as a form of manipulative treatment and described the physiology behind the technique.

A thesis undertaken by Wolfson (4) examined the effectiveness of MET on increasing lumbar flexion. The sample consisted of healthy males and females ages 20-40 who were asymptomatic for low back trouble. The experimental group underwent one treatment of MET with 3 pre-test and 3 post-test measures being taken using the Modified Schober Test. The control group went through the same positioning, but rather than receive the MET, the group was held in a static position. Measurements were taken using the same method as for the experimental group. Results indicated a significant change in lumbar flexion in the experimental group between the last pre-test and first post-test measures. Long term effects were not reported in this study. There may have been an unconscious bias in measurements because the experimenter had knowledge of subject groupings.

Another unpublished study by Harris (5) examined the efficacy of MET on range of motion of the mid cervical spine in a sample of 10 asymptomatic people. This study utilized the combined technique with isometric contraction. A standard goniometer was used as the measuring tool. A total of 6 treatment sessions

were given to the experimental group over a two week period, while the control received no treatments. A *t*-test between pre-test and post-test measures showed no significant change in range of motion in the experimental group. There was, however, a trend toward significance with the mean range of motion measures for the treatment group showing an increase from pre-test to post-test.

Two studies were found on the reliability of the Cervical Range of Motion Device (CROM) which was the measuring tool used for this study.

A study performed by Capuano-Pucci, Rehault, Aukai, Bracke, Day, and Pasterick (8) investigated both intratester and intertester reliability of the cervical range of motion device (CROM) for cervical flexion, extension, lateral flexion, and rotation. The sample consisted of 20 volunteers with no history of cervical pathology.

Pearson product moment correlations for intratester reliability ranged from .63 to .90 for tester 1 and from .62 to .91 for tester 2. Intertester coefficients were found to be good, ranging from .80 to .87 for session number one and .74 to .85 for session number two. Paired differences *t*-tests showed that there were no significant differences between testers or sessions. This suggests the CROM has acceptable intratester and intertester reliability. The authors noted that the population used for this study was healthy individuals and stated that generalizing to patients with cervical dysfunction may not be appropriate.

A second study performed by Rehault, Albright, Byers, Franta, Johnson, Skowronek, and Dougherty (9) was conducted to investigate intertester reliability of the CROM for measuring cervical flexion, extension, lateral flexion, and rotation in subjects with a history of cervical dysfunction. Twenty two subjects were used who had histories including cervical strain or sprain, spondylosis, degenerative disc disease, and undiagnosed neck pain.

Moderate to high intraclass correlation coefficients (ICC = .76 to .98) were found between testers. Mean differences between testers were consistently low for all six measurements (0.5 to 3.6). There was a high reliability for all motions except flexion. These data suggest the CROM is a reliable measuring device between testers for the cervical spine. The authors stated that the guarding effect due to pain or apprehension which occurred at end range may have caused inconsistent measurements between testers.

Methods

The MET procedure as well as the expectations and requirements of being a subject in this study were explained to the volunteers. The experimental design, risks, and benefits were discussed and each volunteer

was asked to read and sign a statement of informed consent before any measurements or procedures were started.

Each person participating was screened to determine if they had a history of cervical pain or pathology by completing a Health History Questionnaire.

Those people who had no history of pathology were then measured for cervical flexion, extension, left and right rotation, and left and right sidebending with the cervical range of motion device (CROM). A combination of techniques from Capuano-Pucci et al. (8) and Rehault et al. (9) was utilized. Each potential subject sat in a straight chair with their feet flat on the ground and their arms at their sides. The subject then looked straight ahead while the CROM was affixed to their ears and bridge of the nose with a velcro strap. The volunteer was then asked to flex, extend, sidebend left, sidebend right, rotate left and rotate right as far as they were able. Each person was instructed to follow a horizontal line for tracking purposes during the rotation components. Persons who had normal or less than a 10 degree restriction in any range of motion were asked not to participate in the study.

Those people who showed a limitation were screened for vertebral artery syndrome. The subject laid supine while the researcher passively extended, sidebent, and rotated the subjects head to one side. The subject then looked into the researcher's eyes and slowly counted to ten while being monitored for nystagmus, dizziness, nausea, and/or light-headedness. Any subject found to show these positive signs was asked not to participate in the study.

Subjects were then randomly placed into either a control or treatment (MET) group. Measurements taken for the inclusion criteria were used for pre-test measures. Each member of the MET group underwent seven treatment sessions over a four week period. Each treatment session consisted of the following procedures as outlined by Greenman (1); The subject was supine. The practitioner then positioned the subject according to their limitation of active range of motion. For example, if the subject had a limitation in extension, left rotation, and left sidebending the practitioner would passively introduce extension, left rotation, and left sidebending to the point of the restriction barrier. Each subject was then asked to produce a small isometric force away from the direction of restriction against the practitioner's hand for approximately five seconds. This procedure was repeated three consecutive times per session. Although each subject was positioned into their individual limitation, the rotation component was resisted for all subjects as all subjects in the MET treatment group presented with limitations in either left or right rotation.

Following the four week treatment program, post-test measurements of cervical flexion, extension,

sidebending, and rotation were taken for both the control and the treatment program using the same technique as described for the pre-test measurements. All subjects in the treatment group had post-test measurements taken one day following their last treatment session.

Results

Description of Sample

Of the 34 volunteers screened, 19 showed a limitation in range of motion. Originally, the control group was made up of nine subjects and the control group was comprised of ten subjects. However, one subject in the control group was unable to attend all treatment sessions, therefore those data were not used. Therefore, the sample was made up of 18 volunteers with nine subjects in the treatment group and nine subjects in the control group. The treatment, or MET group consisted of four males and five females with the mean age of 27. The control group consisted of five males and four females with the mean age of 24.

Not all subjects had similar limitations. Table 1 lists the number of subjects having limitations in each of the six ranges of motion. Note that all subjects had limitations in more than one plane of motion.

Hypothesis

The hypothesis tested in this study stated that a group of subjects with limited range of motion treated with MET would demonstrate a statistically significant increase in cervical flexion, extension, sidebending, and rotation as compared to similar subjects without the MET treatment.

The hypothesis was tested using a series of 6 two-way analysis of variance (ANOVA). The between subjects factor in these analysis was group (MET vs. control), and the within subjects factor was the testing (pre vs. post).

The means for each of the cervical ROM measurements are presented in Table 2 and Figure 1.

Tables 3-8 present the ANOVA summary table for each of the six separate analyses. The key effect

TABLE 1
Number of Subjects Showing Limitation in Each Range of Motion

	MET	Control
Flexion	4	5
Extension	5	4
Left sidebending	2	3
Right sidebending	4	5
Left rotation	3	1
Right rotation	6	4

TABLE 2
Means and Standard Deviations for Cervical Range of Motion in Degrees

		MEAN		SD	
		PRE	POST	PRE	POST
Flexion	MET	51.8	54.4	11.8	6.9
	Control	50.4	49.7	7.0	7.3
Extension	MET	69.1	71.9	10.6	7.9
	Control	71.7	71.7	14.5	12.8
Sidebending					
Left	MET	35.6	40.8	5.3	4.4
	Control	35.9	37.4	7.8	5.2
Right	MET	35.2	43.4	8.4	6.0
	Control	35.0	37.8	3.3	3.6
Rotation					
Left	MET	65.3	73.1	7.3	6.5
	Control	71.8	71.0	8.5	12.5
Right	MET	60.4	69.4	10.4	6.1
	Control	67.4	68.0	11.1	6.2

in each of these analyses is the group by testing interactive effect. Among the six analyses, two ranges of motion had significant interactive effects. These were for left cervical rotation (Table 7, $F(1, 16) = 5.01$ $p = 0.04$) and for right cervical rotation (Table 8, $F(1, 16) = 4.80$ $p = 0.04$). In addition, one range demonstrated a strong trend toward significance. This was for right cervical side bending (Table 6, $F(1, 16) = 4.02$ $p = 0.06$).

Discussion

The subjects in this study did not present with the same limitation in each of the six directions of motion. For example, one subject may have had a limitation in flexion and left rotation while another subject may have had a limitation in extension and right rotation. Each subject was positioned toward their movement restriction. It should be noted that for each subject, rotation was the only direction resisted isometrically. When this factor is compared to the statistical results demonstrated by the series of ANOVA's a conclusion can be drawn that the resisted directions were those which demonstrated statistically significant increases in range of motion. Therefore, individual subjects may have only had a significant change in one direction of motion (the one which was treated). Subsequently, the whole treatment group may not have had significant change in each range of motion. The small number of subjects with identical limitations did not allow statistical analysis of the direction of treatments and how they affected that particular range of motion.

Nonetheless, in each of the six ranges of motion examined in this study, the treatment group demonstrated increased range (post test vs. pre test) while the control group showed little or no change.

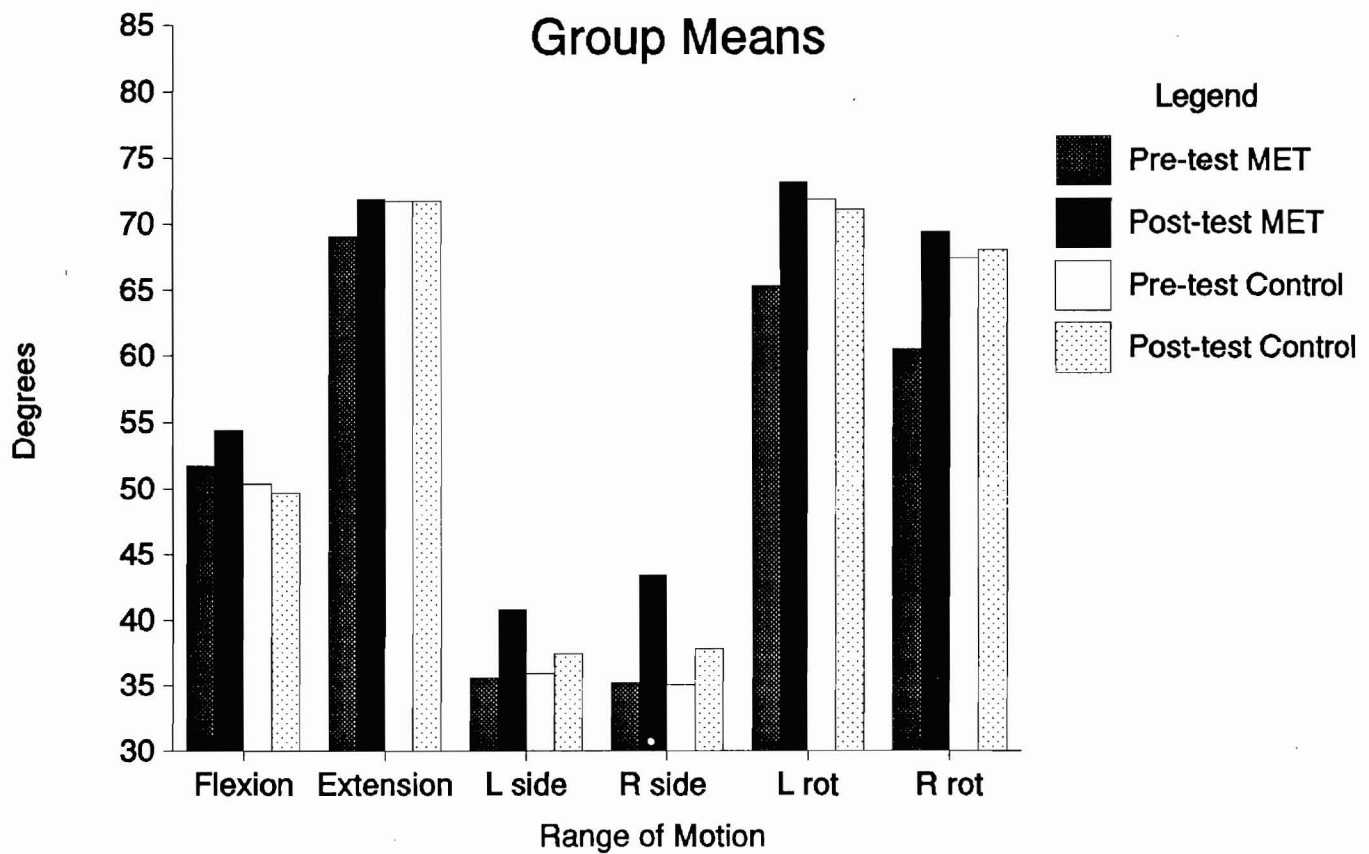


Fig. 1. Cervical range of motion

TABLE 3
ANOVA Summary Table for Flexion

SOURCE	df	SS	MSS	F	P
Between Subjects	17	2088.25			
Group	1	84.03	84.03	0.67	0.42
Error	16	2004.22	125.26		
Within Subjects	18	350.50			
Testing	1	8.03	8.03	0.41	0.53
Group × Testing	1	26.70	26.70	1.35	0.26
Error	16	315.78	19.74		

TABLE 4
ANOVA Summary Table for Extension

SOURCE	df	SS	MSS	F	P
Between Subjects	17	4227.25			
Group	1	12.25	12.25	0.05	0.83
Error	16	4215.00	263.44		
Within Subjects	18	215.50			
Testing	1	17.36	17.36	1.54	0.23
Group × Testing	1	17.36	17.36	1.54	0.23
Error	16	180.78	11.30		

TABLE 5
ANOVA Summary Table for Left Sidebending

SOURCE	df	SS	MSS	F	P
Between Subjects	17	600.25			
Group	1	20.25	20.25	0.56	0.47
Error	16	580.00	36.25		
Within Subjects	18	624.50			
Testing	1	103.36	103.36	3.37	0.09
Group × Testing	1	30.25	30.25	0.99	0.34
Error	16	490.89	30.68		

TABLE 6
ANOVA Summary Table for Right Sidebending

SOURCE	df	SS	MSS	F	P
Between Subjects	17	847.81			
Group	1	78.03	78.03	1.62	0.22
Error	16	624.50	48.11		
Within Subjects	18	604.50			
Testing	1	272.25	272.25	16.40	0.01
Group × Testing	1	66.69	66.69	4.02	0.06
Error	16	265.56	16.60		

TABLE 7
ANOVA Summary Table for Left Rotation

SOURCE	df	SS	MSS	F	P
Between Subjects	17	2109.14			
Group	1	42.25	42.25	0.33	0.58
Error	16	2066.89	129.18		
Within Subjects	18	800.50			
Testing	1	110.25	110.25	3.36	0.09
Group × Testing	1	164.690	164.69	5.01	0.04
Error	16	525.56	32.85		

TABLE 8
ANOVA Summary Table for Right Rotation

SOURCE	df	SS	MSS	F	P
Between Subjects	17	1997.00			
Group	1	69.44	69.44	0.58	0.46
Error	16	1927.56	120.47		
Within Subjects	18	901.00			
Testing	1	205.44	205.44	6.14	0.02
Group × Testing	1	160.44	160.44	4.80	0.04
Error	16	535.11	33.44		

The flexion, extension and side bending movements did not show statistically significant results, however, there were increases in measurements between pre-test and post-test scores for the treatment group in these directions. This would be expected due to the normal arthrokinematics of the cervical spine with the coupling of motions which occurs during movement. This indicates that an increase in one direction of motion would have some carry-over effect to other directions.

Recommendations for Future Research

Post-test measurements for the treatment group were taken one day after the final treatment session. Perhaps measurements would show a greater improvement in range of motion if they had been taken immediately following the final treatment session. Also, there possibly could have been an unconscious bias in measurements by the examiner, who had knowledge of group assignment, however the examiner was not aware of the pre-test measures at the time of the post-test.

A longer treatment period may have produced more significant findings. Similarly, a larger sample size may provide more compelling results.

In this study MET was not performed on the conceptual population of patients who would be seen clinically for limited range of motion. Therefore, future research needs to be done on symptomatic patients with

cervical dysfunction. The results of this study cannot necessarily be generalized to patients having cervical dysfunction.

Research needs to be undertaken comparing MET to other methods of increasing range of motion. These may include passive stretching or active range of motion exercise. MET may be more beneficial than a passive stretching program, or active exercises may increase range of motion just as well as MET. To date, there have been no published empirical studies in this area.

Ideally, this study should have been conducted with all subjects having similar limitations, allowing for subjects to be treated in the same fashion. This extraneous variable may have affected the results.

Conclusion

The MET group demonstrated greater increases compared to the control group in each of the six ranges of motion measured in this study. Even though, a strong trend toward significance was noted in right side bending, only right and left rotation were found to demonstrate statistical significance. The lack of significance in other ranges may be attributed to the subjects presenting with heterogeneous cervical limitations. In addition, the MET group was treated toward left and right rotation.

These results suggest that MET is an effective method to increase cervical range of motion.

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